

CE2007

Microprocessor-based System Design

Memory Technologies

Contents

Characteristics and operating principles of

- Flash Memory
- SRAM – Static Random Access Memory
- DRAM – Dynamic Random Access Memory

Introduction

Memory devices are used to store program and data that microprocessor executes.

Different types of memory devices are used to meet different requirements.

In particular, solid state memories (i.e. non-mechanical type) are increasingly being used in microprocessor based systems of today

- primarily due to speed and power consumption, as well as reliability



Memory Classification

Classified into two main types

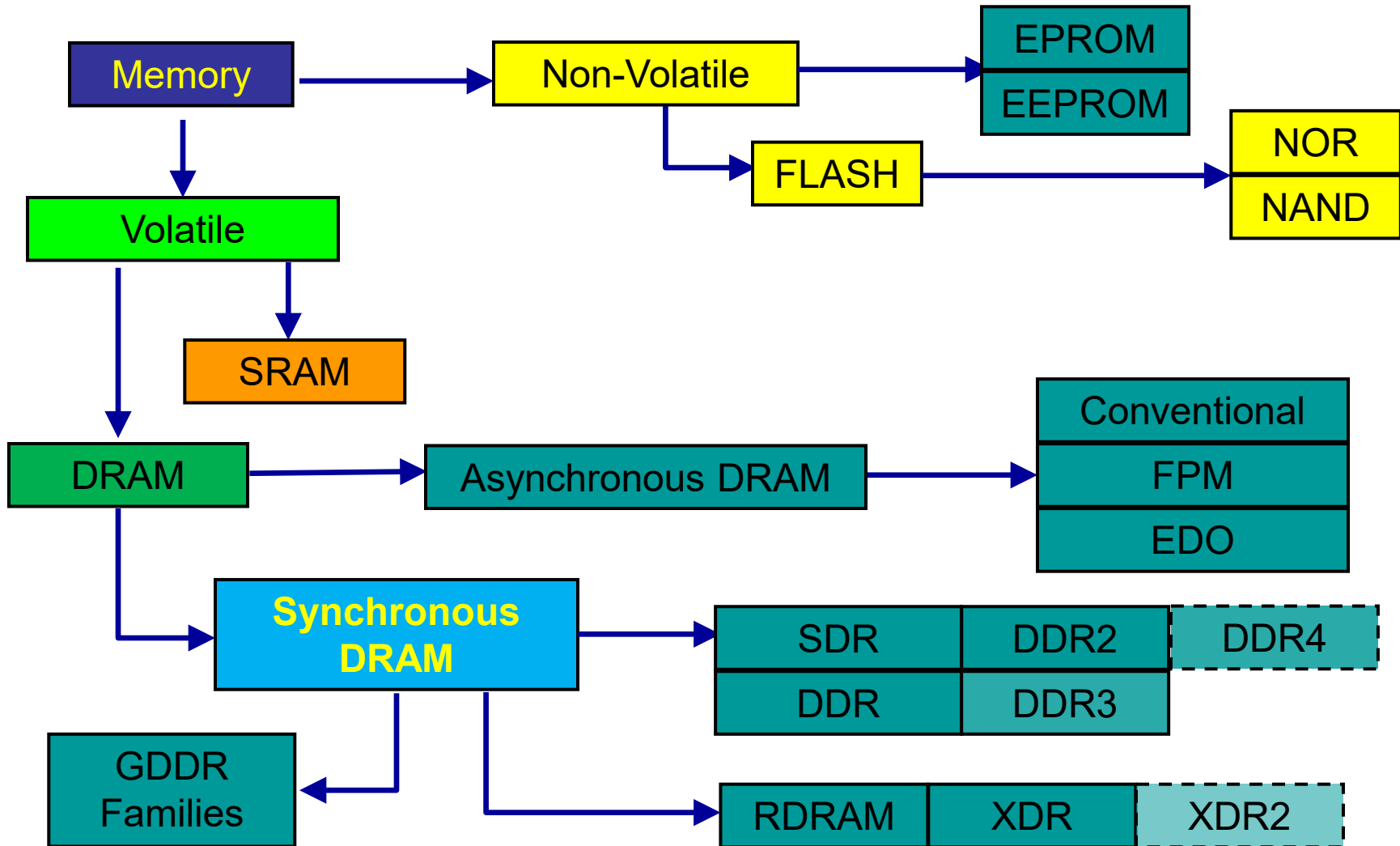
1. Non-Volatile

- doesn't need power to retain data
- useful for storing boot up firmware and storing data

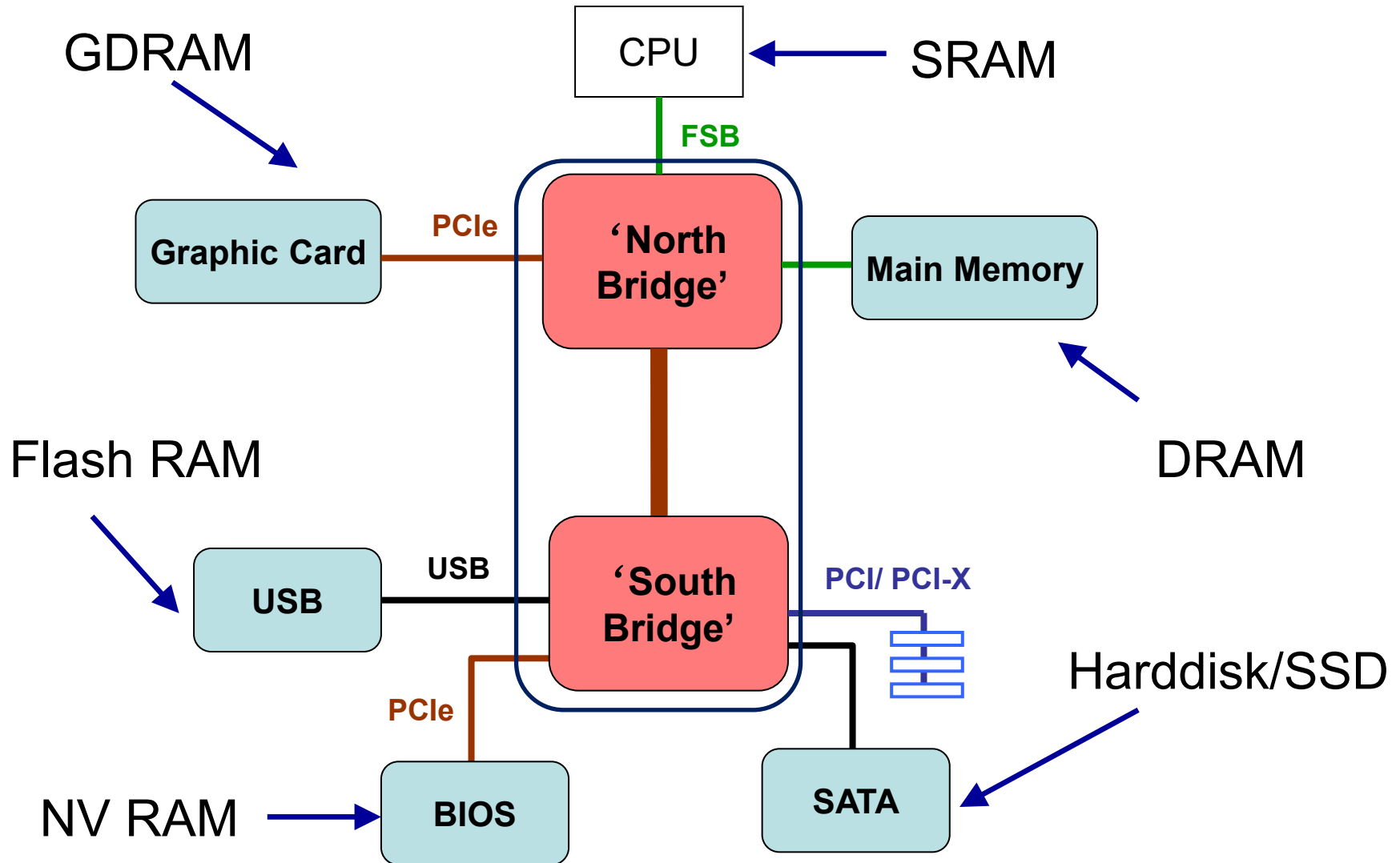
2. Volatile

- needs continuous power to retain data
- much faster than the non-volatile memory
- used in PC's cache memory and main memory

Memory Device Families



Memory Used in PC System



Non Volatile Memory

Non-Volatile Memory

Can retain data even when there is no power applied

- for program/firmware storage
- store Look-up tables
- store configuration data for embedded system when it is in-circuit programmable
- for data logging and collection (Data acquisition)
- for data storage if there is no storage drive like harddisk. (Digital Photos, MP3 music, thumb Drive, SSD drive)

Non-Volatile Memory

Types of Non-Volatile memory

- ROM - Read only Memory
- PROM – Programmable Read Only Memory
- EPROM - Erasable Programmable Read Only
Memory



(The above are largely obsolete)

- EEPROM - Electrically Erasable Programmable Read
Only Memory
- FLASH Memory

FLASH Memory Technologies

FLASH Memory

Main features

- can be erased in a 'flash'
- can be erased and programmed in-socket/on-board
- less expensive and higher capacities than EEPROM
 - but more 'complicated' to program
 - contain on-chip 'controller'
 - need to issue a series of 'commands' to program
- random Read Access time in the order of sub-hundreds nanoseconds
- usually Block oriented for write access (write in block/sector/page)
- often used in embedded systems that need to be reprogrammed and updated in the field
- but now?

FLASH Types

Two types of Flash Memory available

- conventional NOR type
- 'newer' but now well established NAND type

NOR type

- faster Read Access time than NAND type
- useful for executing code directly (e.g. for mobile phone)

NAND type

- higher density (1Gbits vs. 128Mbits) and faster write
- useful for storing of data
 - USB memory stick, flash card for digital camera and multimedia player, SD card for smart phone/tablet
 - SSD

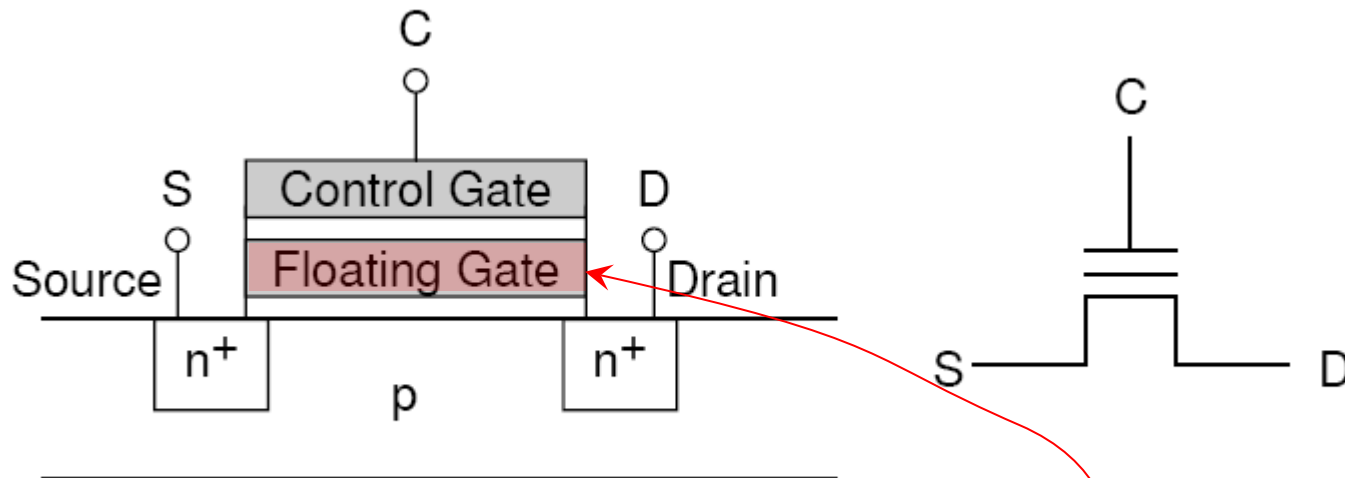
Review Question 1

- Which of the following memories are suitable for storing the BIOS in your computer?

(There can be more than one answer)

- A. EEPROM
- B. SRAM
- C. NAND Flash
- D. NOR Flash
- E. DRAM

Reading FLASH Memory Cell



To read the contents of the cell

- a voltage is placed across CS

Current flowing through DS

- **depends on the charge stored** in the Floating Gate.
- no current if the floating gate has (sufficient) charges

By measuring I_{DS}

- charges in the Floating Gate is found.

Programming FLASH Memory Cell

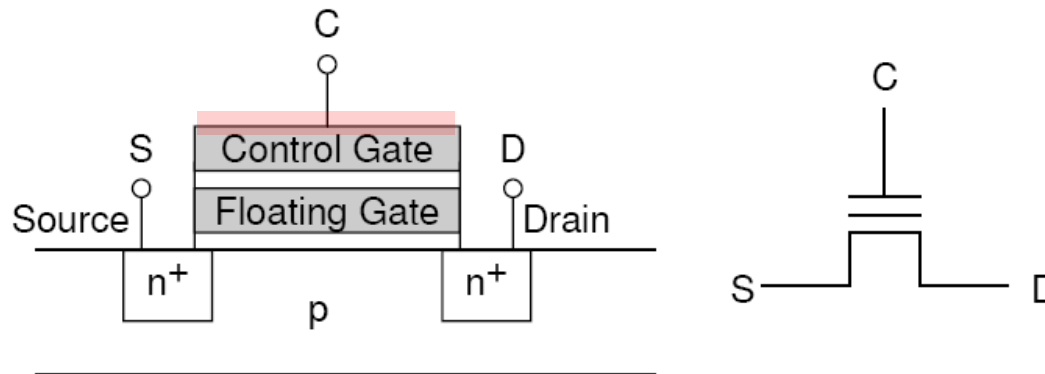
To program a '0' to the cell

- simultaneously pulsing the Drain with 5V and the Control gate with 10V;
- charge is then stored on the floating gate

To program a '1' to the cell (e.g. erase)

- simultaneously pulsing the Source with 5V and the Control gate with -8V;
- charge is then removed

Hence to re-program a memory cell, cell must first be erased



Flash (Erase) Cycle

A NAND flash cell faces extremely high electrical field

- needed to force the electrons to pass through the oxide layer in either direction.

Over time, the tunnel oxide layer is degraded

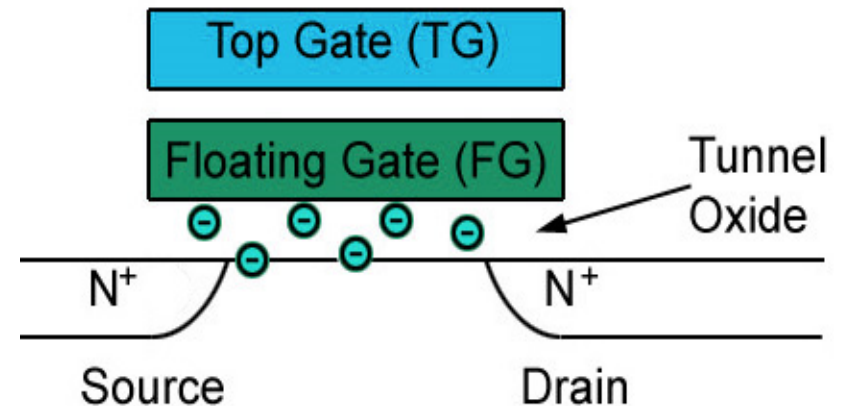
- result is trapping of electrons in the layer
- erase process becomes slower whereas the programming becomes 'faster'

Eventually, enough charges is accumulated in the oxide layer

- cell can no longer be erased (i.e. reset to "1")

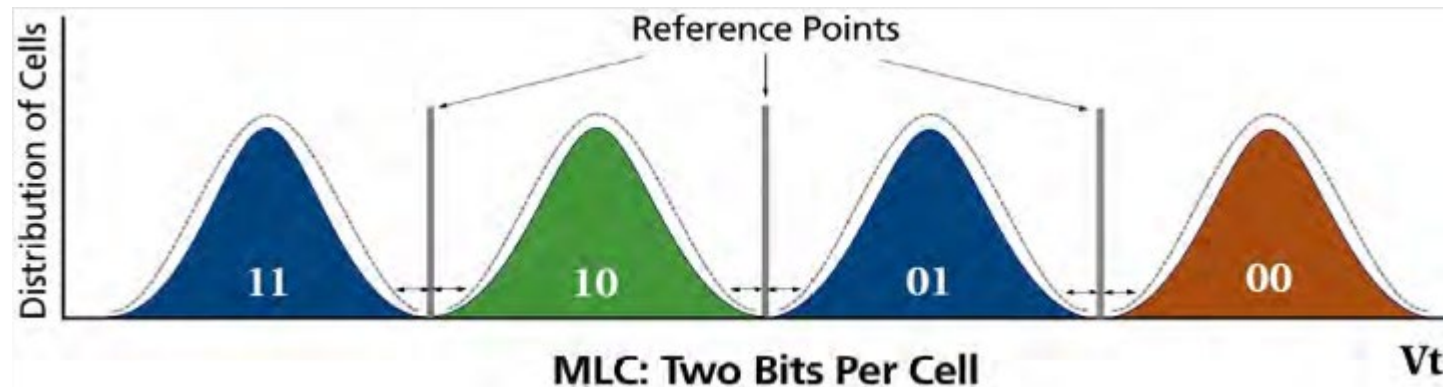
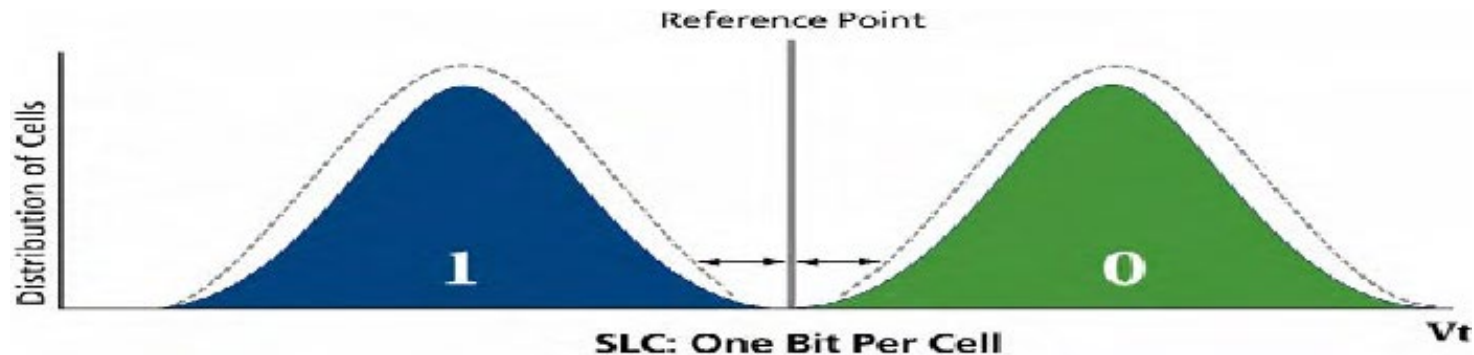
Maximum limit to erase cycle per cell

- in the range of 10 000 to 1000 000
- need a 'smart' firmware/driver to spread the erase-write operation among different cells
e.g. using a wear-leveling technique



Multi-Level Cell

Multi-level cell (MLC) Flash technology enable higher storage density than traditional single-level cell (SLC) Flash.



Volatile Memory

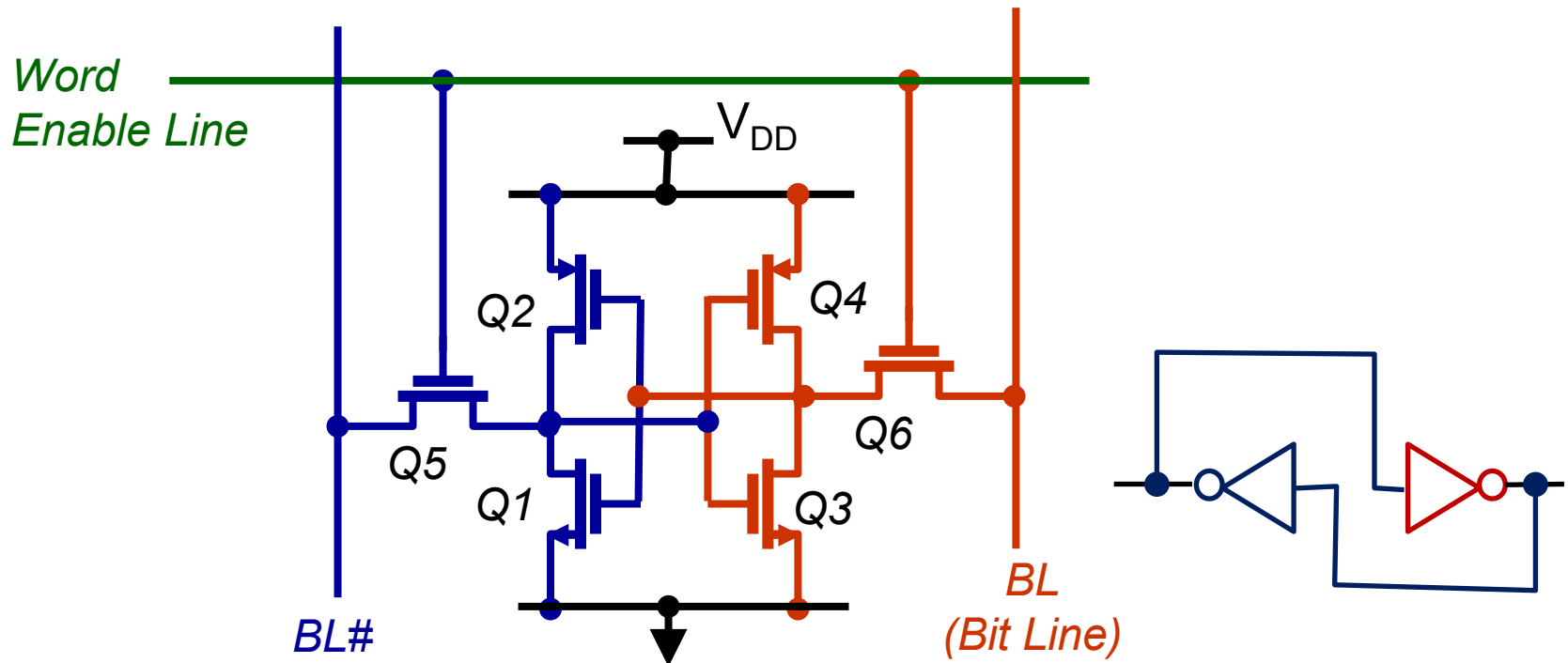
SRAM

(Static Random Access Memory)

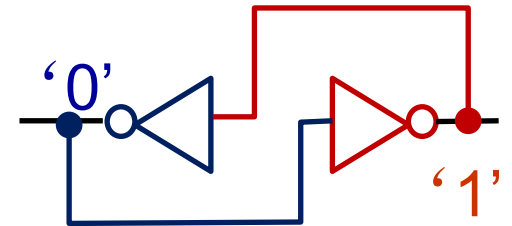
SRAM Cell

A 6-MOS transistors version

- two cross coupled inverters which form a latch
- differential driven bit lines for reliability

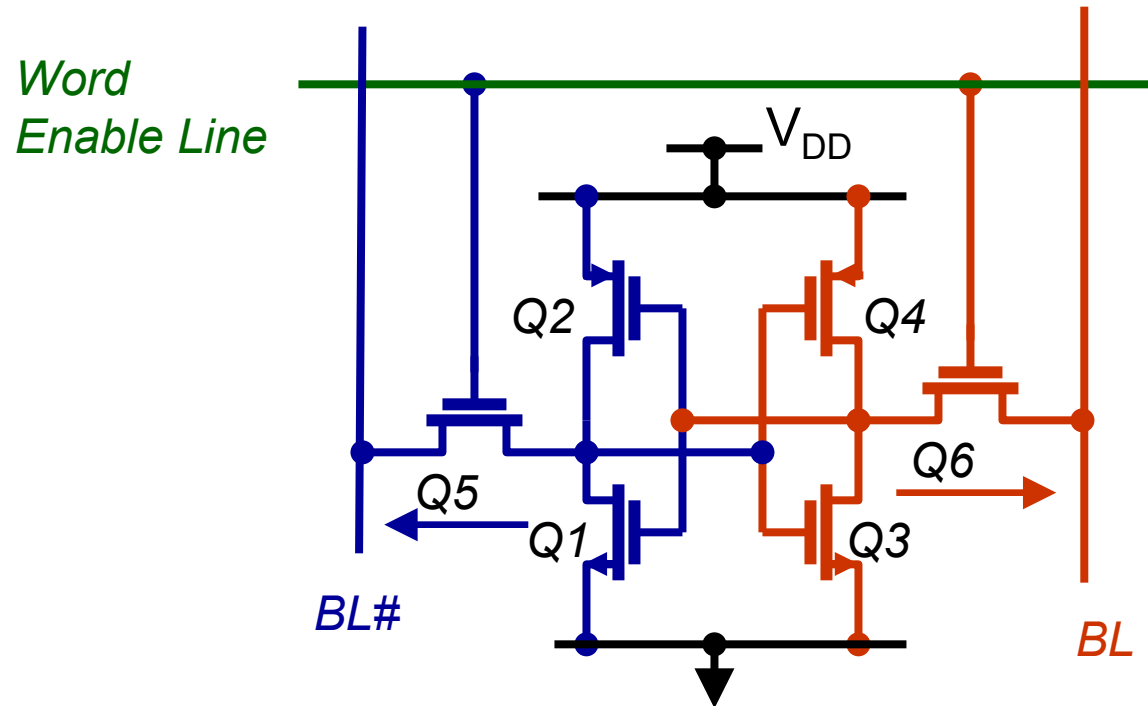


*Word
Enable Line*



- output of each CMOS inverter pair reinforced each other
- data is retained indefinitely as long as voltage is applied

Read Operation



BL & BL# are first **precharged** to HIGH

- *Word Enable* line is then asserted to select the cell
- Either BL & BL# will be pulled LOW by the cell
- a **Sense Amplifier** detects the differential output between the two lines to output logical value of '1' or '0'

SRAM Characteristics

SRAM memory cell is basically made of D Latches

- can retain data as long as power(voltage) is applied
- no periodic refresh needed (as that done in DRAM)
 - hence fast access time of 10 nsec and less

SRAM

- fastest memory available
 - access time can be in a few nanoseconds
- used as expansion memory for microcontroller based embedded systems due to simplicity in interfacing
- used as main memory for portable device for its low power consumption
- used as cache memory (together with DRAM as main memory) for high performance (speed) systems

DRAM

(Dynamic Random Access Memory)

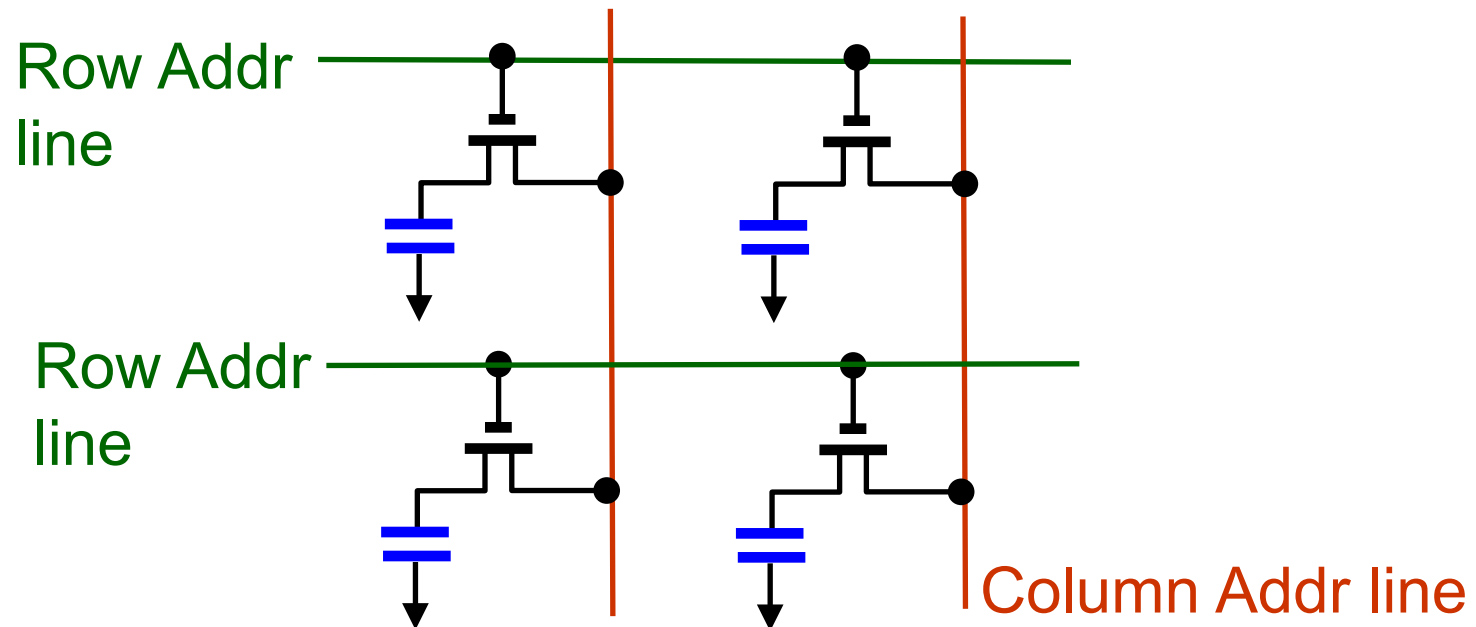
DRAM Cells

Dynamic Random Accessed Memory (DRAM)

- consists of memory 'cells'

Each memory cell

- made of one MOS Transistor and one capacitor
- arranged in a grid of Row-Column fashion
- accessed through selecting the 'row' and the 'column'



Cell Access

Number of rows and columns in a DRAM

- generally equal to a power of two

Example:

A one Megabit DRAM that has 1024x1024 memory cells

- uses 10-bit row and 10-bit column address lines.

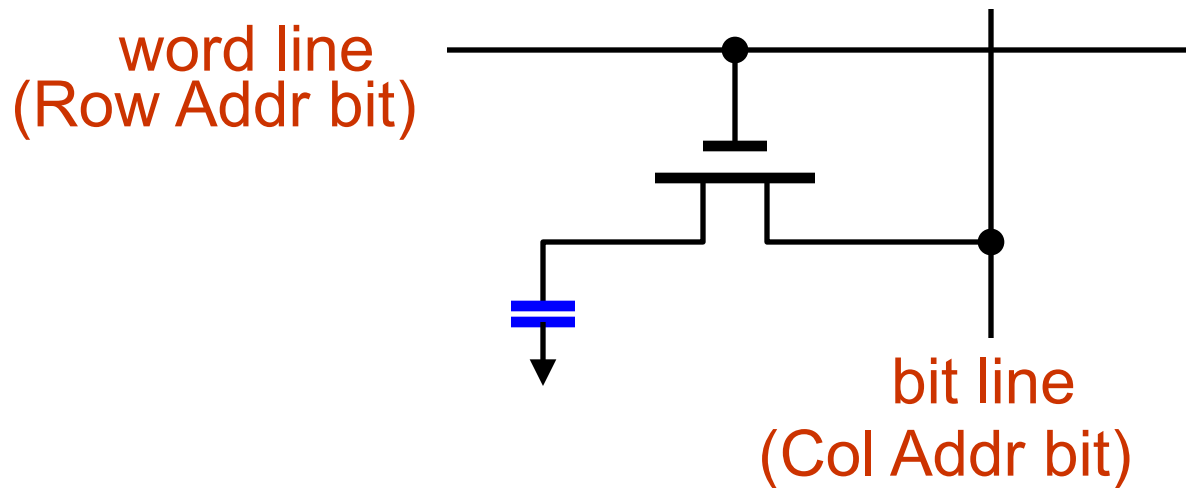
To access a cell

- first select the row address (**Word Line**)
(i.e. 1024 cells will be selected at the same time)
- then select the column address (**Bit line**)
which specify the particular cell

DRAM Cell

Each DRAM memory cell

- consists of one MOS Transistor and one capacitor
- data is retained as charge in the capacitor



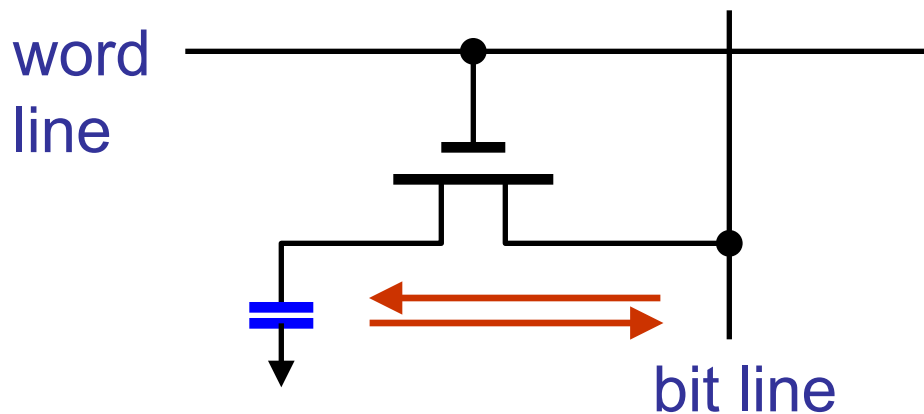
Typical value of capacitance used

- in the region of few tens of fF (10^{-15}F)
e.g. 30fF for 50nm process technology

Writing to a DRAM Cell

To write to a cell

1. Value is first output onto the bit line
2. Word line is then asserted
 - charge or discharge the capacitor



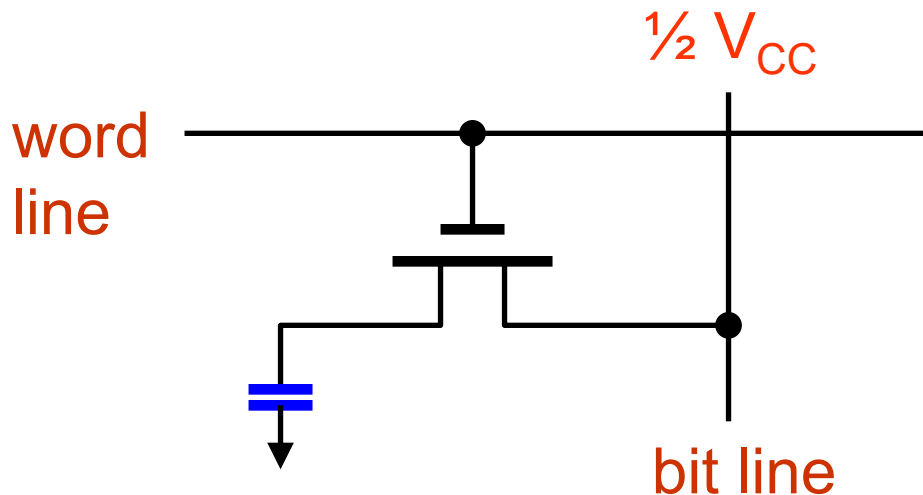
DRAM Cell Read Operation

To get a cell ready for (read) access

1. Bit line is first **precharged** to a reference voltage halfway between logic HIGH and LOW

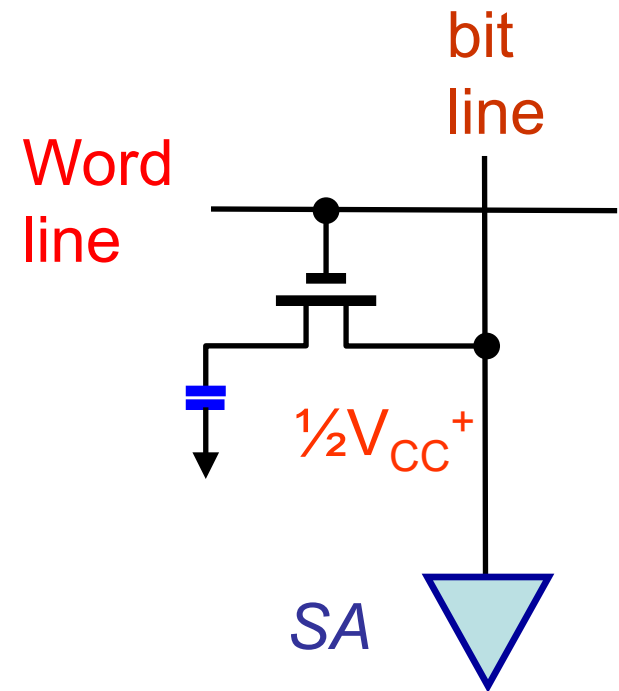
In practice, precharge of the bit lines is typically performed after a cell has been accessed on the line

- equivalent to the reset of the bit line
- get ready for next access of the memory



DRAM Cell Read Operation

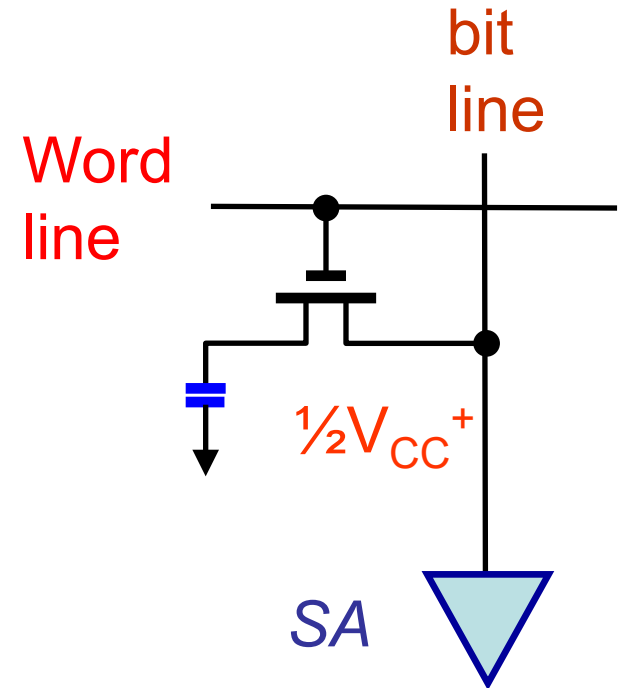
2. **Word line** is asserted (driven HIGH) to connect cell's storage capacitor to its bit-line
 - Pre-charged bit line is pulled slightly higher or lower (a bit-line typically has much more capacitance than a cell capacitor)
3. **Sense Amplifier** is used to detect the small change (e.g. 200mV) in the bit line
 - Output the corresponding logic value



DRAM Cell Read Operation

Note that the process of reading a cell 'disturbs' the charges of the cell

4. Restore the charges by writing back the data after reading (done automatically by SA)
5. Word-line is switched off to disconnect the storage cell capacitors from the bit-lines. The sense amplifier is switched off, and the bit lines are pre-charged again.



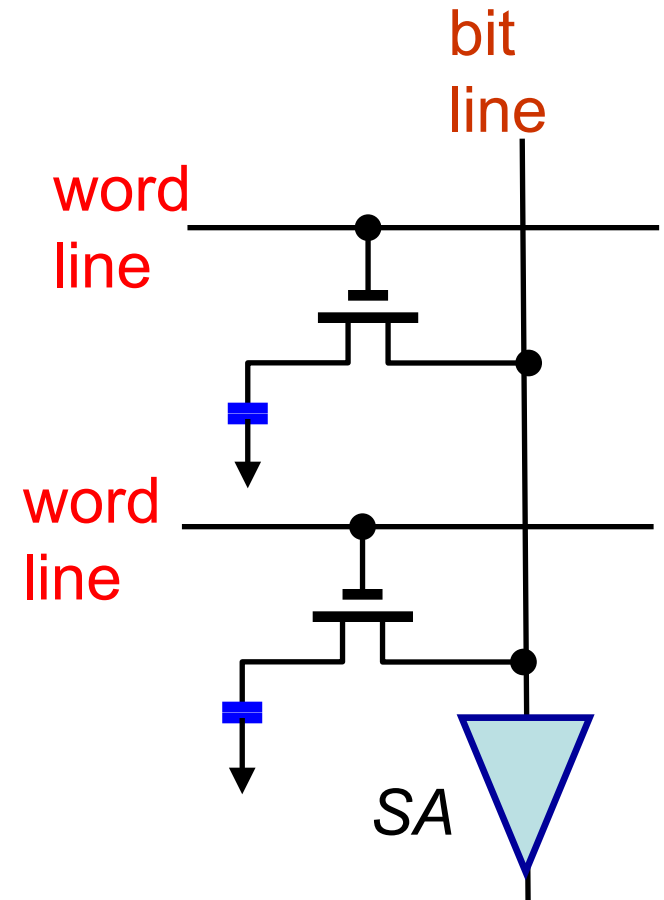
Sense Amplifier

Sensing of bit line

- is done differentially
- slowest process during the accessing of the memory cells

Sense Amplifier

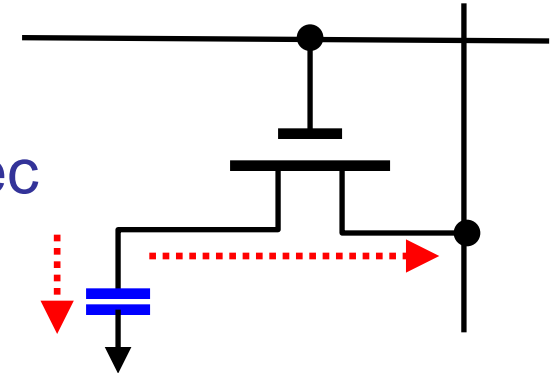
- occupies more area than cell
- consumes additional current when holding data
- one unit shared for each bit line



DRAM Cell Refresh

DRAM cell's capacitor is also discharged

- due to leakage current
- data retained for several hundreds msec
- need to refresh each cell periodically



Refresh Cycle involves

- **reading** of each cell content and store into a latch
- **writing back** a 'solid' logic value from the latch

Rate of refresh needed to ensure cells are not discharged

- usually performed at a rate of $15.6 \mu\text{s}$ interval (per device)
- actual refresh duration is less than 100 ns

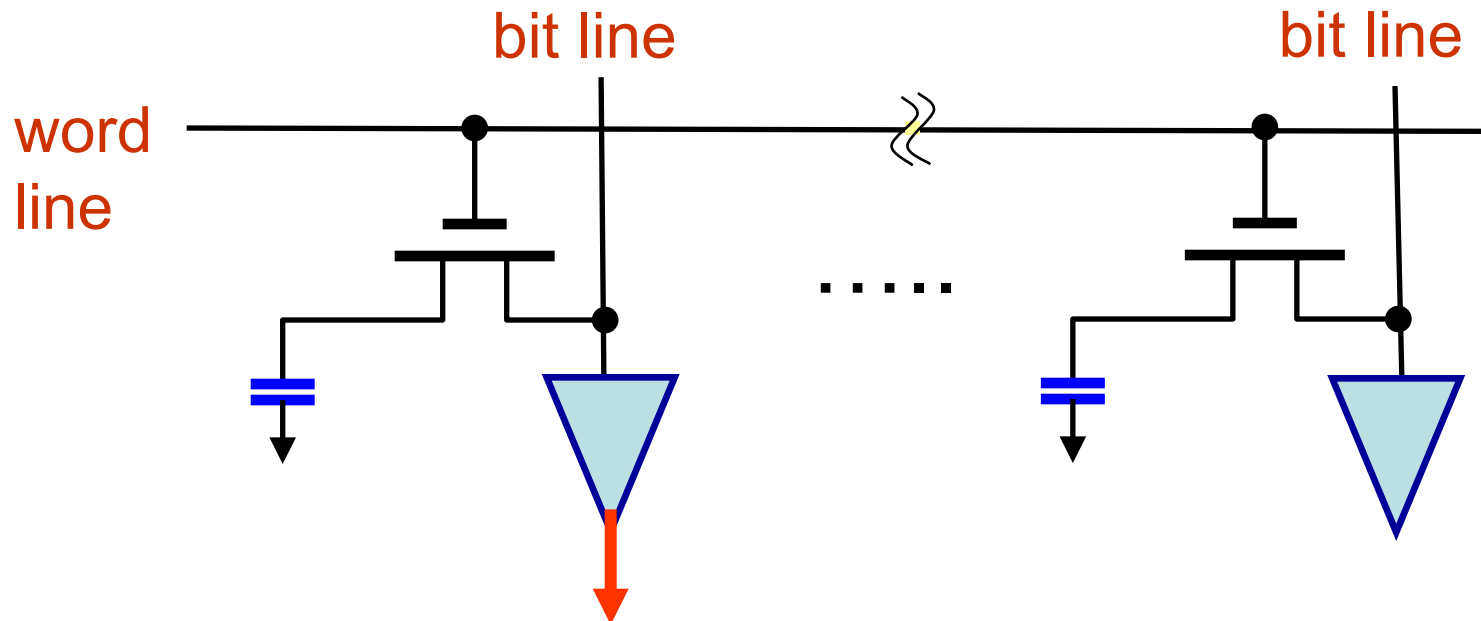
DRAM Access

Due to structure of DRAM cells arrangement

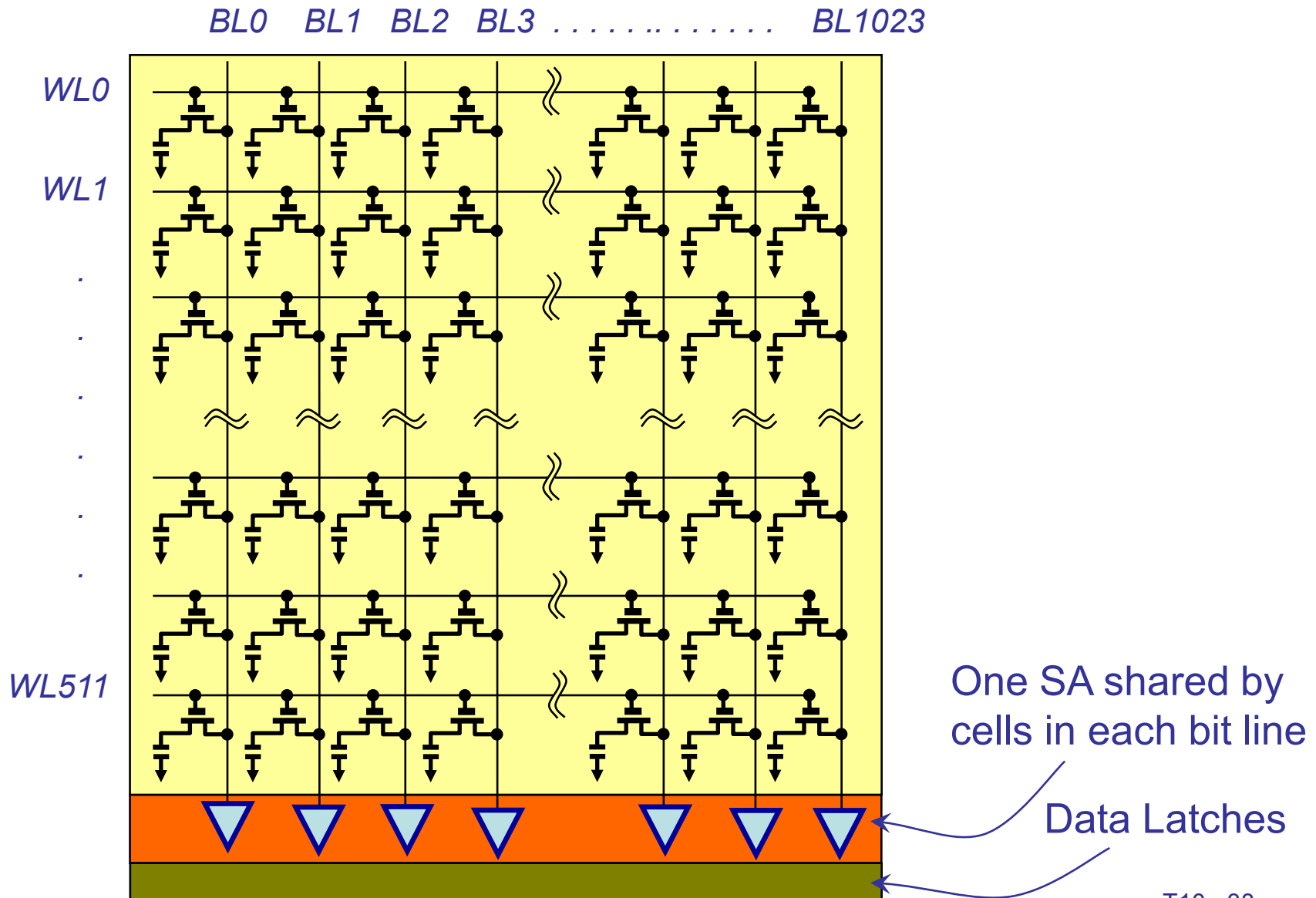
- access of DRAM is performed **one whole row** at a time

During Read

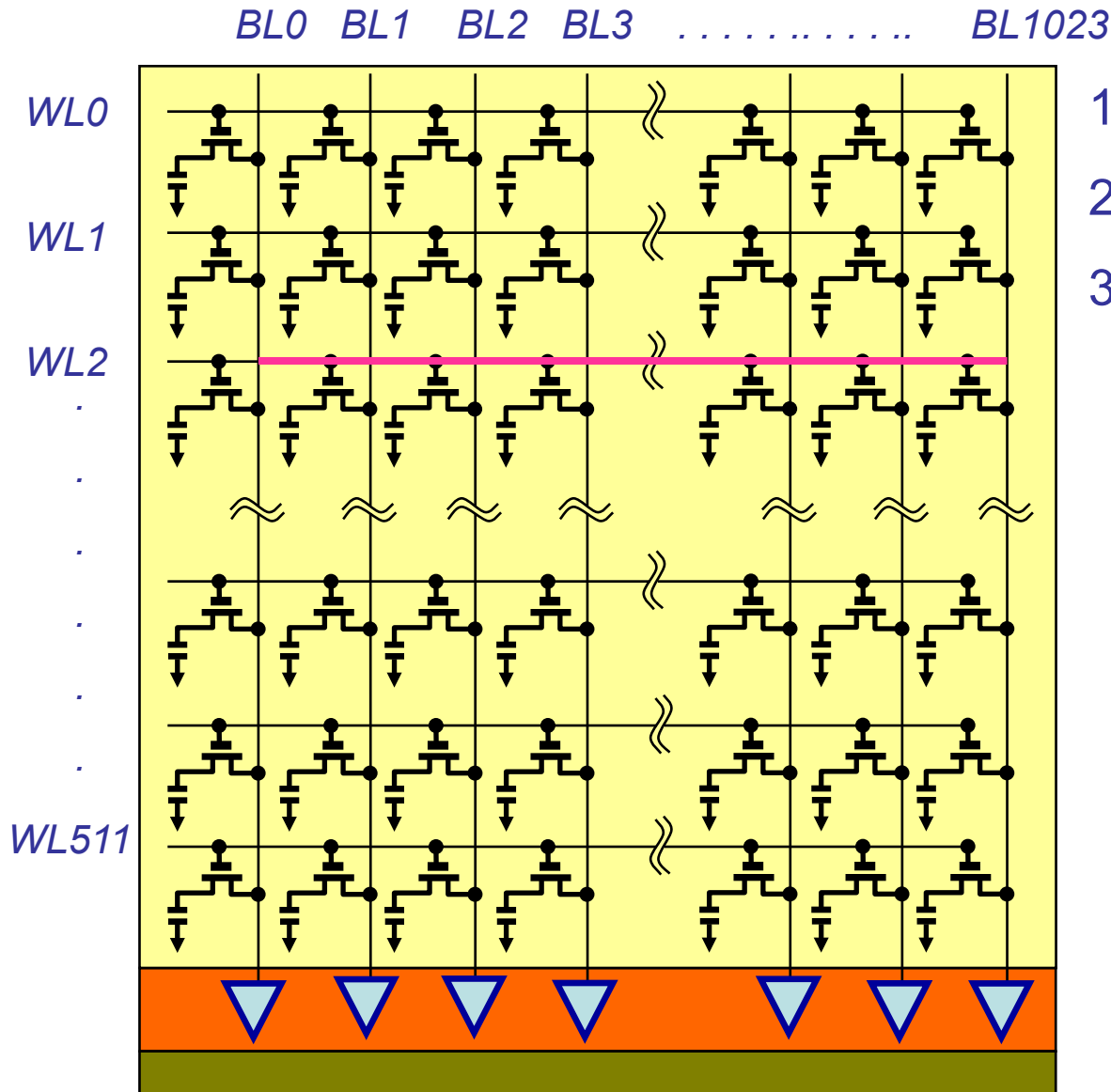
- row selected is output to the all bit lines
- but only the selected bit line's SA is eventually output to the data bus



A 512 x 1024 1-bit Memory Cells Array

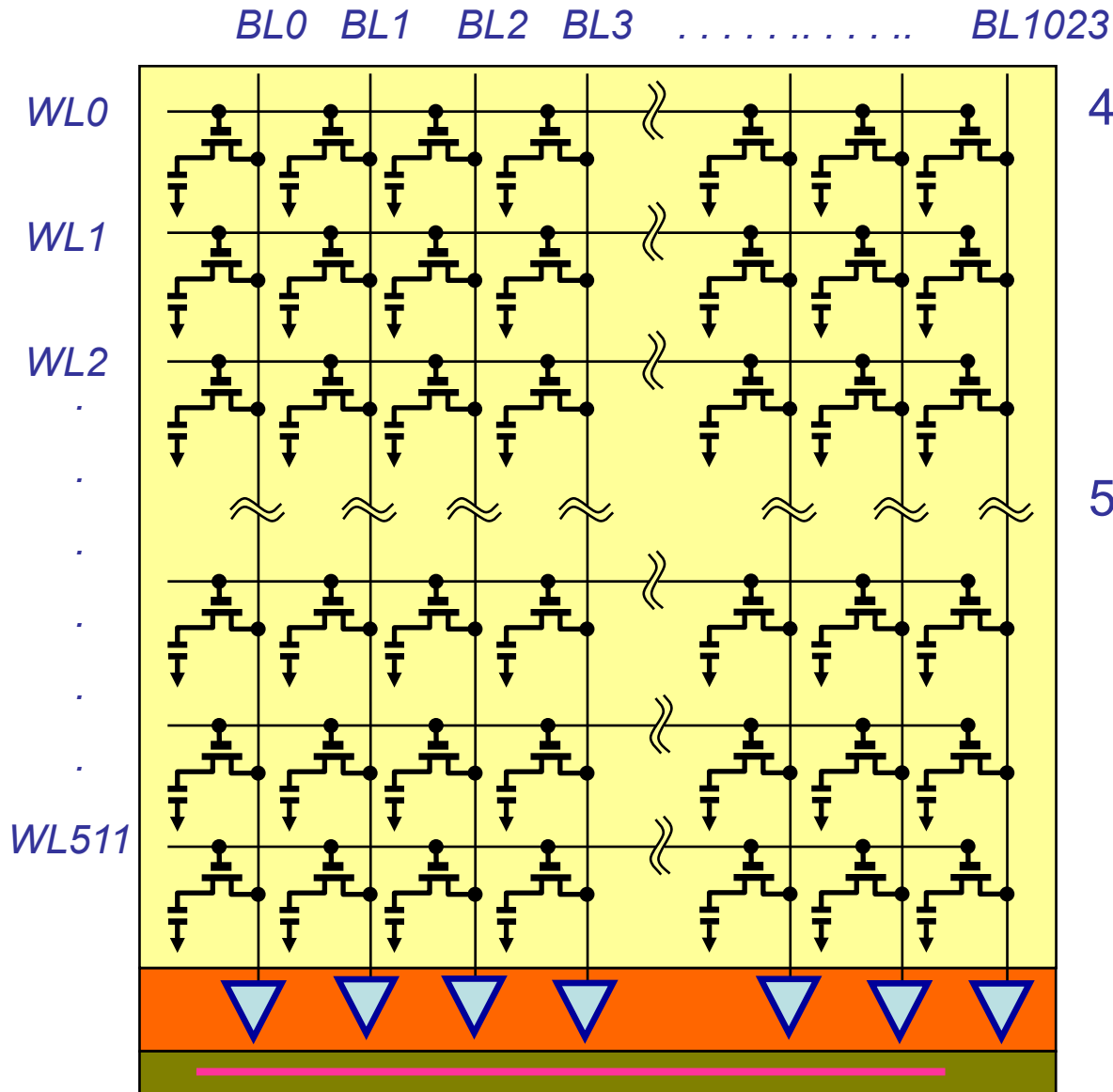


DRAM Read Access



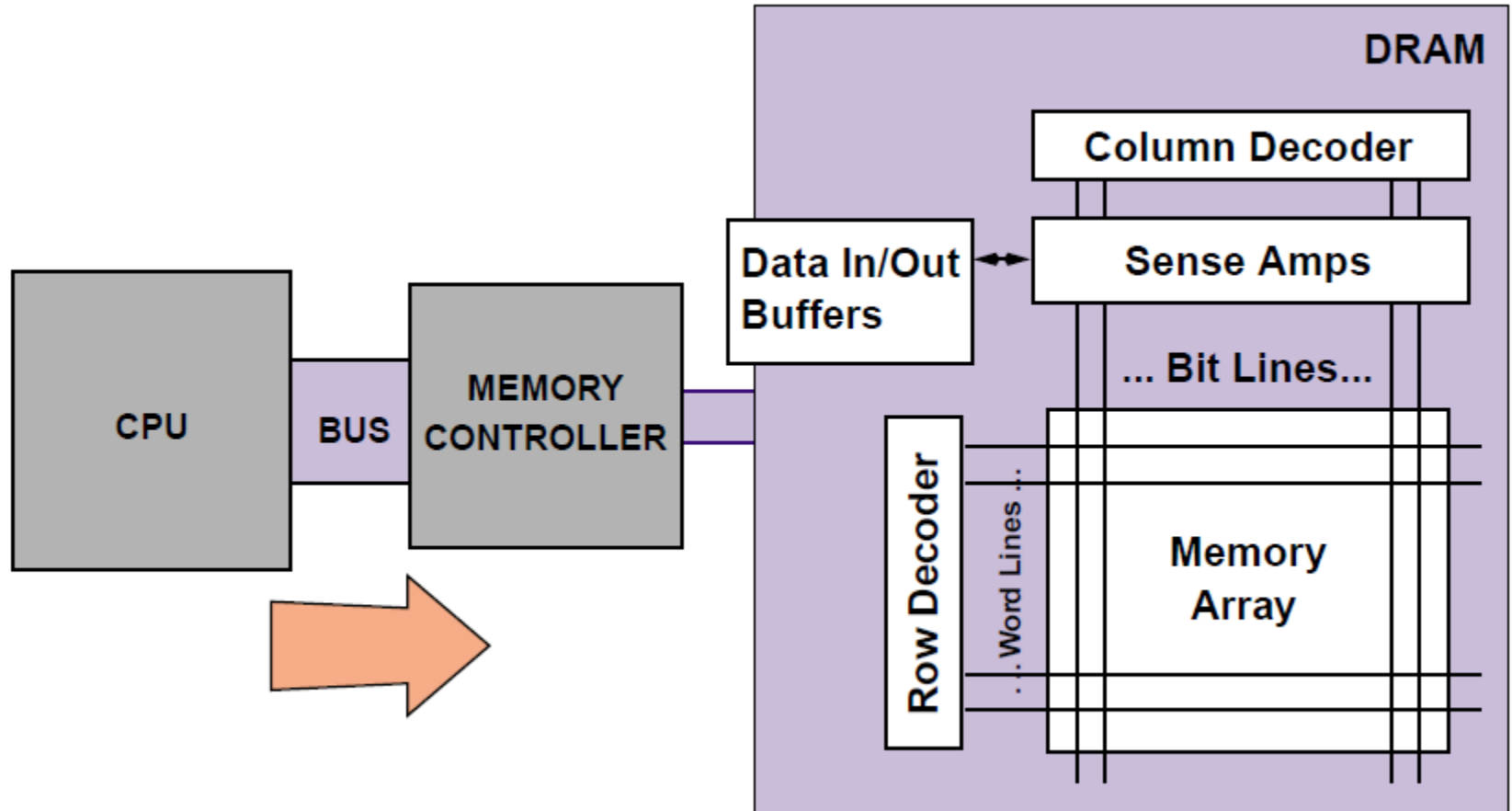
- 1) Precharge Bit Lines
- 2) Select Word line
- 3) All storage cells in the open row are sensed simultaneously, and the sense amplifier outputs latched. A column address then selects which latch bit to connect to the external data bus.

DRAM Read Access



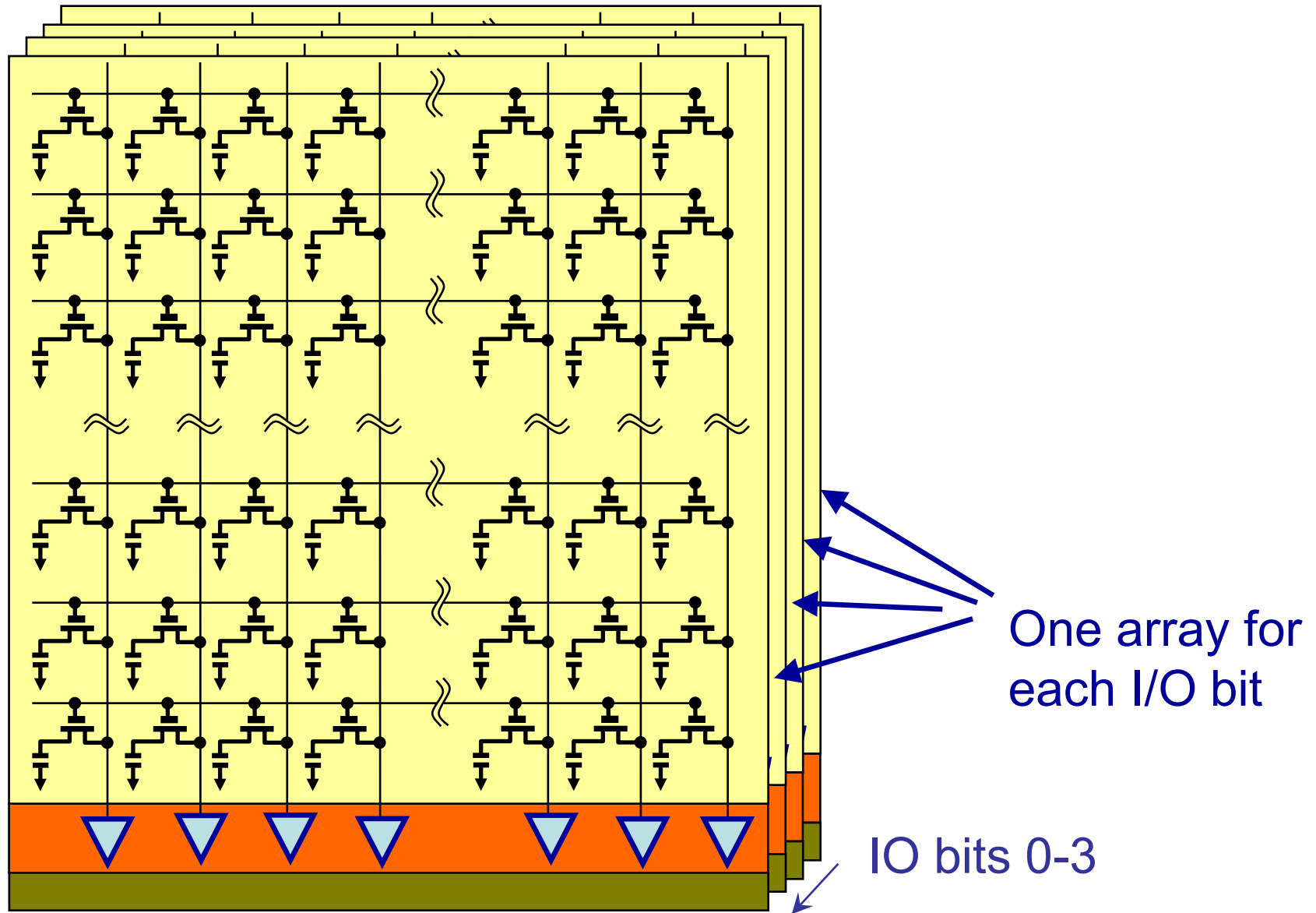
- 4) While reading of columns, current flows back up the bit-lines from the output of the sense amplifiers and recharge storage cells
- 5) Word-line and sense amplifier is switched off, and the bit lines are pre-charged again

DRAM Interface

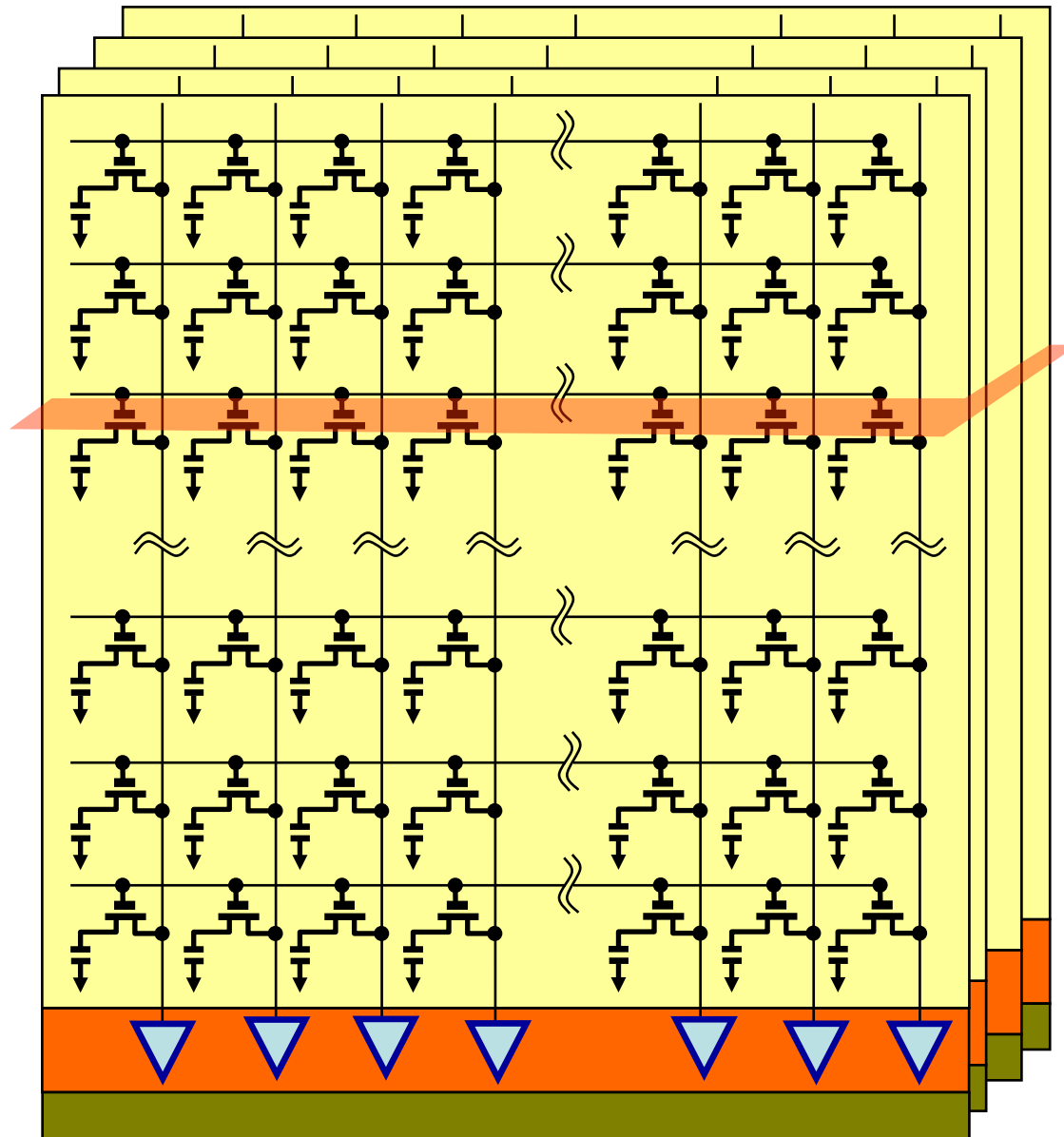


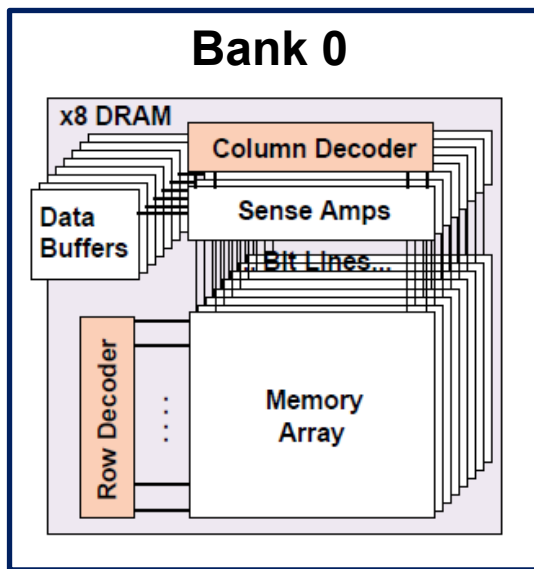
Source: Bruce et. al. - ISCA 2002

4-bit Memory Array

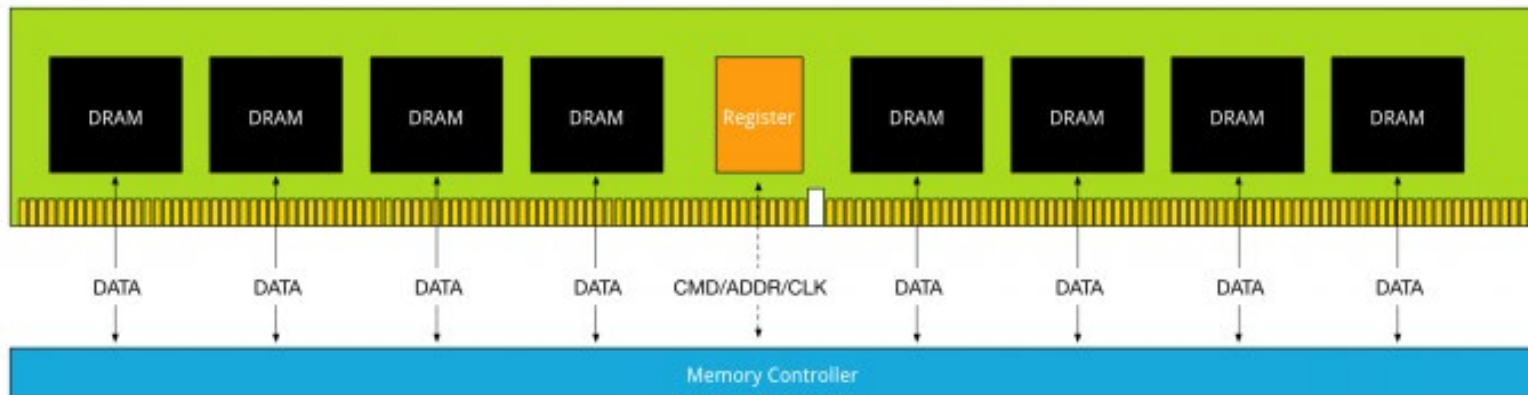
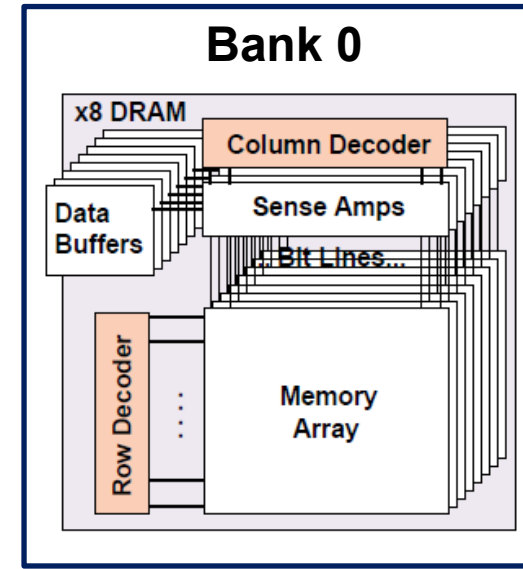


Opening a 'Page'





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Source: <http://frankdenneman.nl/2015/02/25/memory-deep-dive-ddr4/>



SDRAM

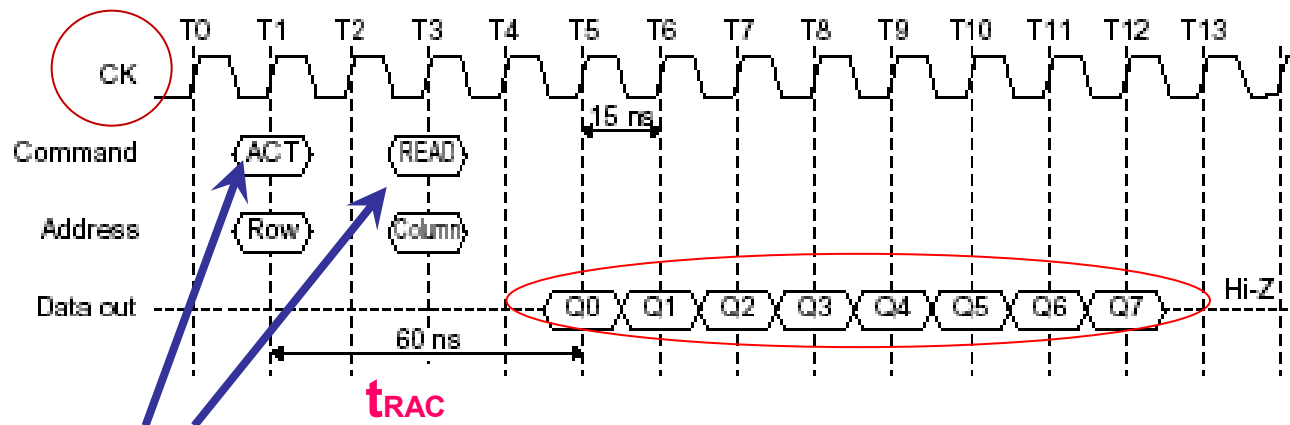
SDRAM uses fast **synchronous clock** interface

Uses **command driven** interface

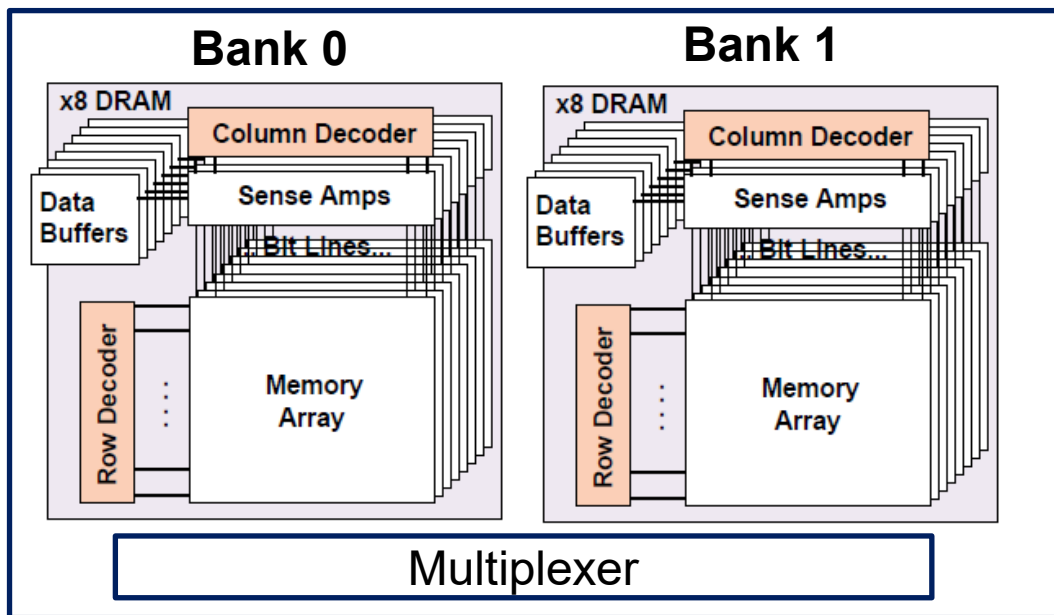
- encoded using RAS, CAS, CS & WE signals

Allows **burst read access (consecutive data per clock)**

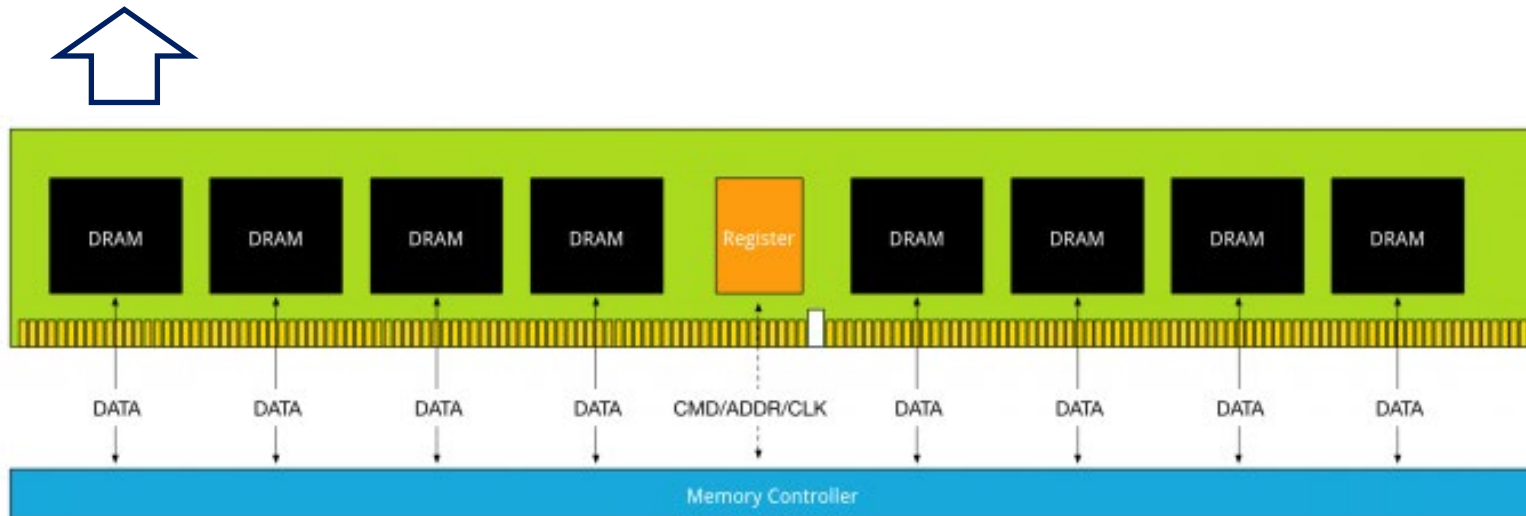
- internal burst counter increments the column address
- amortize the effect of 1st random access delay t_{RAC}



Commands (based on RAS, CAS, CS, nWE combinations)

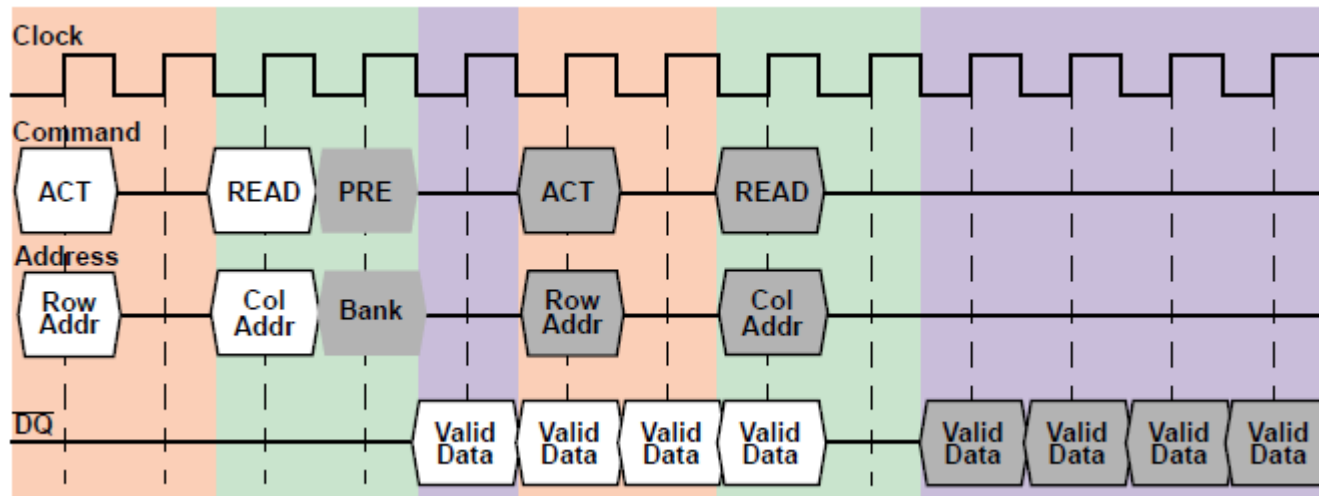
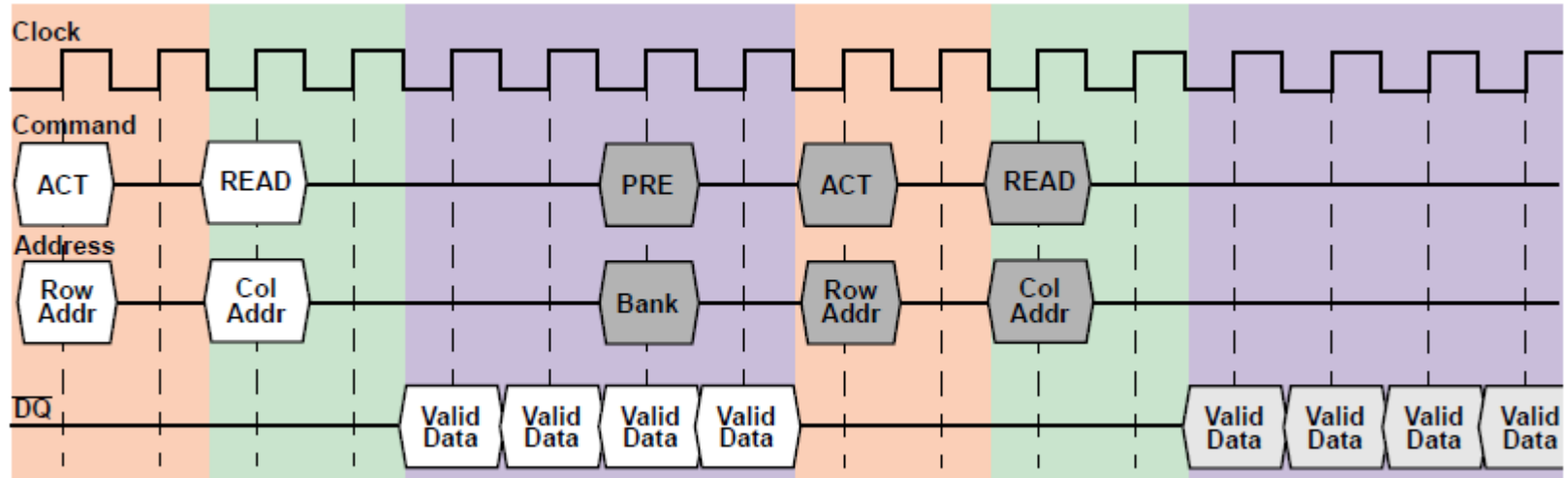


Why 2 Banks?



Source: <http://frankdenneman.nl/2015/02/25/memory-deep-dive-ddr4/>

Access to DDR-SDRAM, Bank Overlap



Access to another bank can already start when the first burst is trasmitted.

Variations of DRAM

DRAM are available in a number of variations

- but with **internal memory cell's operation remains similar**
- difference lies in the interface

Asynchronous (DRAM) – Obsolete

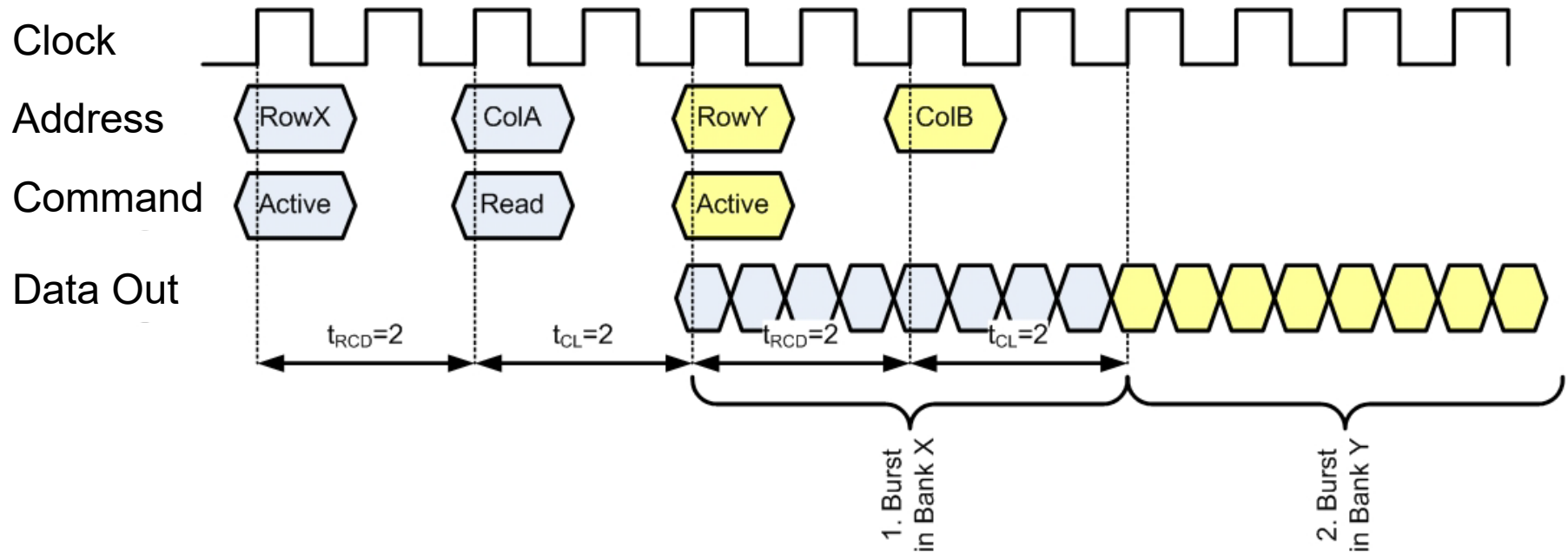
- a) Conventional (original)
- b) FPM (Fast Page Mode)
- c) EDO (Extended Data Out)

Synchronous (**SDRAM**)

- a) SDR (Single Data Rate)
- b) DDR (Double Data Rate)
- c) DDR2, DDR3, DDR4

Other examples: Rambus RDRAM, XDR, Graphic GDDRx

Access to DDR-SDRAM, Bank Overlap



Access to another bank can already start when the first burst is transmitted.

DRAM vs. SRAM

DRAM

- more cost effective per bit
 - has higher density per silicon area than SRAM
- but needs periodic refresh cycles to maintain data
 - slower than SRAM
- multiplexed address lines (row/column)
 - smaller packaging size
 - but more complicated interfacing

Need memory controller to issue commands, multiplexed row/column address and issue refresh cycles.

DRAM vs. SRAM (cont)

SRAM

- no refresh is needed
 - more power efficient
 - much faster as cell and data is always ready and available
 - no refresh controller needed
- interface is simpler
 - no multiplexing of row and column address
 - no memory controller needed to issue commands
- use more transistor per cell
 - lower bit density